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Source: *Journal of the Royal Statistical Society. Series B (Methodological)*, 1996, Vol. 58, No. 1 (1996), pp. 267-288

Published by: Oxford University Press for the Royal Statistical Society

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J. R. Statist. Soc. B (1996)
58, *No.* 1, *pp.* 267–288

Regression Shrinkage and Selection via the Lasso

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[Received January 1994. Revised January 1995]

SUMMARY

We propose a new method for estimation in linear models. The 'lasso' minimizes the residual sum of squares subject to the sum of the absolute value of the coefficients being less than a constant. Because of the nature of this constraint it tends to produce some coefficients that are exactly 0 and hence gives interpretable models. Our simulation studies suggest that the lasso enjoys some of the favourable properties of both subset selection and ridge regression. It produces interpretable models like subset selection and exhibits the stability of ridge regression. There is also an interesting relationship with recent work in adaptive function estimation by Donoho and Johnstone. The lasso idea is quite general and can be applied in a variety of statistical models: extensions to generalized regression models and tree-based models are briefly described.

Keywords: QUADRATIC PROGRAMMING; REGRESSION; SHRINKAGE; SUBSET SELECTION

1. INTRODUCTION

Consider the usual regression situation: we have data $(\mathbf{x}^i, y_i)$, $i = 1, 2, \ldots, N$, where $\mathbf{x}^i = (x_{i1}, \ldots, x_{ip})^T$ and y_i are the regressors and response for the ith observation. The ordinary least squares (OLS) estimates are obtained by minimizing the residual squared error. There are two reasons why the data analyst is often not satisfied with the OLS estimates. The first is *prediction accuracy*: the OLS estimates often have low bias but large variance; prediction accuracy can sometimes be improved by shrinking or setting to 0 some coefficients. By doing so we sacrifice a little bias to reduce the variance of the predicted values and hence may improve the overall prediction accuracy. The second reason is *interpretation*. With a large number of predictors, we often would like to determine a smaller subset that exhibits the strongest effects.

The two standard techniques for improving the OLS estimates, subset selection and ridge regression, both have drawbacks. Subset selection provides interpretable models but can be extremely variable because it is a discrete process—regressors are either retained or dropped from the model. Small changes in the data can result in very different models being selected and this can reduce its prediction accuracy. Ridge regression is a continuous process that shrinks coefficients and hence is more stable: however, it does not set any coefficients to 0 and hence does not give an easily interpretable model.

We propose a new technique, called the *lasso*, for 'least absolute shrinkage and selection operator'. It shrinks some coefficients and sets others to 0, and hence tries to retain the good features of both subset selection and ridge regression.

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In Section 2 we define the lasso and look at some special cases. A real data example is given in Section 3, while in Section 4 we discuss methods for estimation of prediction error and the lasso shrinkage parameter. A Bayes model for the lasso is briefly mentioned in Section 5. We describe the lasso algorithm in Section 6. Simulation studies are described in Section 7. Sections 8 and 9 discuss extensions to generalized regression models and other problems. Some results on soft thresholding and their relationship to the lasso are discussed in Section 10, while Section 11 contains a summary and some discussion.

2. THE LASSO

2.1. *Definition*

Suppose that we have data $(\mathbf{x}^i, y_i)$, $i = 1, 2, \ldots, N$, where $\mathbf{x}^i = (x_{i1}, \ldots, x_{ip})^T$ are the predictor variables and y_i are the responses. As in the usual regression set-up, we assume either that the observations are independent or that the y_is are conditionally independent given the x_{ij}s. We assume that the x_{ij} are standardized so that $\Sigma_i x_{ij}/N = 0$, $\Sigma_i x_{ij}^2/N = 1$.

Letting $\hat{\beta} = (\hat{\beta}_1, \ldots, \hat{\beta}_p)^T$, the lasso estimate $(\hat{\alpha}, \hat{\beta})$ is defined by

$$(\hat{\alpha}, \hat{\beta}) = \arg\min\left\{\sum_{i=1}^{N}\left(y_i - \alpha - \sum_j \beta_j x_{ij}\right)^2\right\} \quad \text{subject to} \sum_j |\beta_j| \leqslant t. \tag{1}$$

Here $t \geqslant 0$ is a tuning parameter. Now, for all t, the solution for α is $\hat{\alpha} = \bar{y}$. We can assume without loss of generality that $\bar{y} = 0$ and hence omit α.

Computation of the solution to equation (1) is a quadratic programming problem with linear inequality constraints. We describe some efficient and stable algorithms for this problem in Section 6.

The parameter $t \geqslant 0$ controls the amount of shrinkage that is applied to the estimates. Let $\hat{\beta}_j^o$ be the full least squares estimates and let $t_0 = \Sigma|\hat{\beta}_j^o|$. Values of $t < t_0$ will cause shrinkage of the solutions towards 0, and some coefficients may be exactly equal to 0. For example, if $t = t_0/2$, the effect will be roughly similar to finding the best subset of size $p/2$. Note also that the design matrix need not be of full rank. In Section 4 we give some data-based methods for estimation of t.

The motivation for the lasso came from an interesting proposal of Breiman (1993). Breiman's *non-negative garotte* minimizes

$$\sum_{i=1}^{N}\left(y_i - \alpha - \sum_j c_j \hat{\beta}_j^o x_{ij}\right)^2 \quad \text{subject to } c_j \geqslant 0, \quad \sum c_j \leqslant t. \tag{2}$$

The garotte starts with the OLS estimates and shrinks them by non-negative factors whose sum is constrained. In extensive simulation studies, Breiman showed that the garotte has consistently lower prediction error than subset selection and is competitive with ridge regression except when the true model has many small non-zero coefficients.

A drawback of the garotte is that its solution depends on both the sign and the magnitude of the OLS estimates. In overfit or highly correlated settings where the OLS estimates behave poorly, the garotte may suffer as a result. In contrast, the lasso avoids the explicit use of the OLS estimates.

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Frank and Friedman (1993) proposed using a bound on the L^q-norm of the parameters, where q is some number greater than or equal to 0; the lasso corresponds to $q = 1$. We discuss this briefly in Section 10.

2.2. *Orthonormal Design Case*

Insight about the nature of the shrinkage can be gleaned from the orthonormal design case. Let $\mathbf{X}$ be the $n \times p$ design matrix with ijth entry x_{ij}, and suppose that $\mathbf{X}^T\mathbf{X} = \mathbf{I}$, the identity matrix.

The solutions to equation (1) are easily shown to be

$$\hat{\beta}_j = \text{sign}(\hat{\beta}_j^o)(|\hat{\beta}_j^o| - \gamma)^+ \tag{3}$$

where γ is determined by the condition $\Sigma|\hat{\beta}_j| = t$. Interestingly, this has exactly the same form as the soft shrinkage proposals of Donoho and Johnstone (1994) and Donoho *et al.* (1995), applied to wavelet coefficients in the context of function estimation. The connection between soft shrinkage and a minimum L_1-norm penalty was also pointed out by Donoho *et al.* (1992) for non-negative parameters in the context of signal or image recovery. We elaborate more on this connection in Section 10.

In the orthonormal design case, best subset selection of size k reduces to choosing the k largest coefficients in absolute value and setting the rest to 0. For some choice of λ this is equivalent to setting $\hat{\beta}_j = \hat{\beta}_j^o$ if $|\hat{\beta}_j^o| > \lambda$ and to 0 otherwise. Ridge regression minimizes

$$\sum_{i=1}^{N} \left(y_i - \sum_j \beta_j x_{ij}\right)^2 + \lambda \sum_j \beta_j^2$$

or, equivalently, minimizes

$$\sum_{i=1}^{N} \left(y_i - \sum_j \beta_j x_{ij}\right)^2 \qquad \text{subject to} \sum \beta_j^2 \leqslant t. \tag{4}$$

The ridge solutions are

$$\frac{1}{1+\gamma}\hat{\beta}_j^o$$

where γ depends on λ or t. The garotte estimates are

$$\left(1 - \frac{\gamma}{\hat{\beta}_j^{o2}}\right)^+ \hat{\beta}_j^o.$$

Fig. 1 shows the form of these functions. Ridge regression scales the coefficients by a constant factor, whereas the lasso translates by a constant factor, truncating at 0. The garotte function is very similar to the lasso, with less shrinkage for larger coefficients. As our simulations will show, the differences between the lasso and garotte can be large when the design is not orthogonal.

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2.3. *Geometry of Lasso*

It is clear from Fig. 1 why the lasso will often produce coefficients that are exactly 0. Why does this happen in the general (non-orthogonal) setting? And why does it not occur with ridge regression, which uses the constraint $\Sigma \beta_j^2 \leqslant t$ rather than $\Sigma|\beta_j| \leqslant t$? Fig. 2 provides some insight for the case $p = 2$.

The criterion $\Sigma_{i=1}^{N}(y_i - \Sigma_j \beta_j x_{ij})^2$ equals the quadratic function

$$(\beta - \hat{\beta}^o)^T \mathbf{X}^T \mathbf{X} (\beta - \hat{\beta}^o)$$

(plus a constant). The elliptical contours of this function are shown by the full curves in Fig. 2(a); they are centred at the OLS estimates; the constraint region is the rotated square. The lasso solution is the first place that the contours touch the square, and this will sometimes occur at a corner, corresponding to a zero coefficient. The picture for ridge regression is shown in Fig. 2(b): there are no corners for the contours to hit and hence zero solutions will rarely result.

An interesting question emerges from this picture: can the signs of the lasso estimates be different from those of the least squares estimates $\hat{\beta}_j^o$? Since the variables are standardized, when $p = 2$ the principal axes of the contours are at $\pm 45°$ to the co-ordinate axes, and we can show that the contours must contact the square in the same quadrant that contains $\hat{\beta}^o$. However, when $p > 2$ and there is at least moderate correlation in the data, this need not be true. Fig. 3 shows an example in three dimensions. The view in Fig. 3(b) confirms that the ellipse touches the constraint region in an octant different from the octant in which its centre lies.

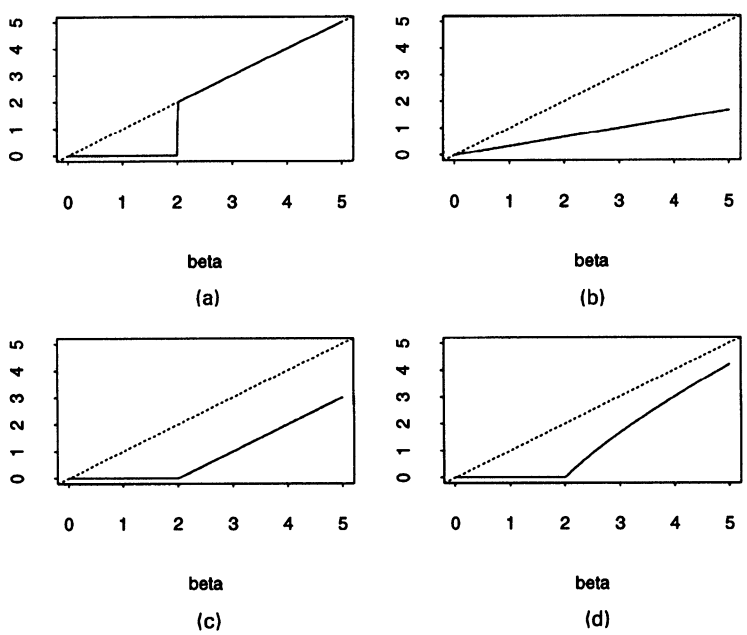


Fig. 1. (a) Subset regression, (b) ridge regression, (c) the lasso and (d) the garotte: ———, form of coefficient shrinkage in the orthonormal design case; ·····, 45°-line for reference

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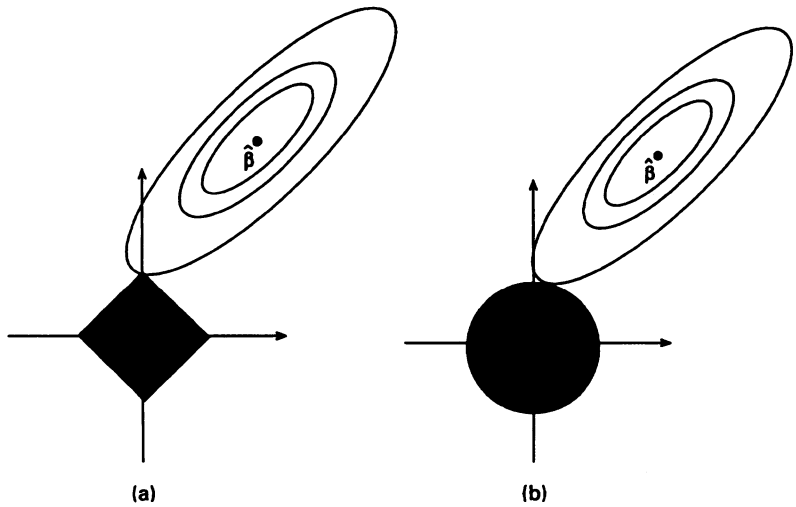


Fig. 2. Estimation picture for (a) the lasso and (b) ridge regression

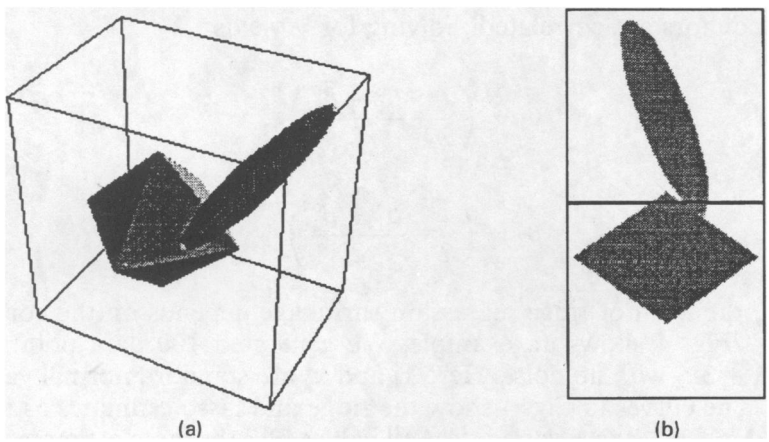


Fig. 3. (a) Example in which the lasso estimate falls in an octant different from the overall least squares estimate; (b) overhead view

Whereas the garotte retains the sign of each $\hat{\beta}_j^o$, the lasso can change signs. Even in cases where the lasso estimate has the same sign vector as the garotte, the presence of the OLS estimates in the garotte can make it behave differently. The model $\Sigma c_j \hat{\beta}_j^o x_{ij}$ with constraint $\Sigma c_j \leqslant t$ can be written as $\Sigma \beta_j x_{ij}$ with constraint $\Sigma \beta_j / \hat{\beta}_j^o \leqslant t$. If for example $p = 2$ and $\hat{\beta}_1^o > \hat{\beta}_2^o > 0$ then the effect would be to stretch the square in Fig. 2(a) horizontally. As a result, larger values of β_1 and smaller values of β_2 will be favoured by the garotte.

2.4. *More on Two-predictor Case*

Suppose that $p = 2$, and assume without loss of generality that the least squares estimates $\hat{\beta}_j^o$ are both positive. Then we can show that the lasso estimates are

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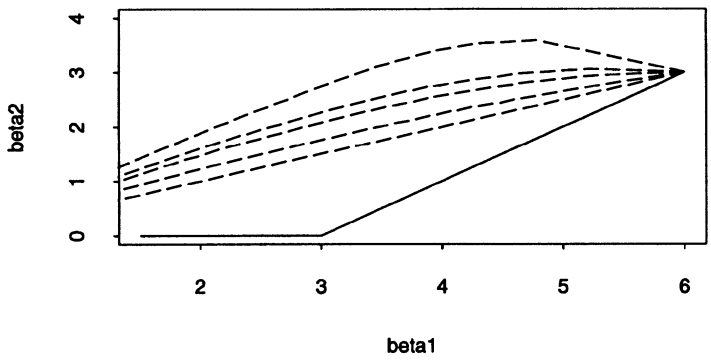


Fig. 4. Lasso (———) and ridge regression (– – – –) for the two-predictor example: the curves show the (β_1, β_2) pairs as the bound on the lasso or ridge parameters is varied; starting with the bottom broken curve and moving upwards, the correlation ρ is 0, 0.23, 0.45, 0.68 and 0.90

$$\hat{\beta} = (\hat{\beta}_j^{\circ} - \gamma)^+ \tag{5}$$

where γ is chosen so that $\hat{\beta}_1 + \hat{\beta}_2 = t$. This formula holds for $t \leqslant \hat{\beta}_1^{\circ} + \hat{\beta}_2^{\circ}$ and is valid even if the predictors are correlated. Solving for γ yields

$$\begin{aligned} \hat{\beta}_1 &= \left(\frac{t}{2} + \frac{\hat{\beta}_1^{\circ} - \hat{\beta}_2^{\circ}}{2}\right)^+, \\ \hat{\beta}_2 &= \left(\frac{t}{2} - \frac{\hat{\beta}_1^{\circ} - \hat{\beta}_2^{\circ}}{2}\right)^+. \end{aligned} \tag{6}$$

In contrast, the form of ridge regression shrinkage depends on the correlation of the predictors. Fig. 4 shows an example. We generated 100 data points from the model $y = 6x_1 + 3x_2$ with no noise. Here x_1 and x_2 are standard normal variates with correlation ρ. The curves in Fig. 4 show the ridge and lasso estimates as the bounds on $\beta_1^2 + \beta_2^2$ and $|\beta_1| + |\beta_2|$ are varied. For all values of ρ the lasso estimates follow the full curve. The ridge estimates (broken curves) depend on ρ. When $\rho = 0$ ridge regression does proportional shrinkage. However, for larger values of ρ the ridge estimates are shrunken differentially and can even increase a little as the bound is decreased. As pointed out by Jerome Friedman, this is due to the tendency of ridge regression to try to make the coefficients equal to minimize their squared norm.

2.5. *Standard Errors*

Since the lasso estimate is a non-linear and non-differentiable function of the response values even for a fixed value of t, it is difficult to obtain an accurate estimate of its standard error. One approach is via the bootstrap: either t can be fixed or we may optimize over t for each bootstrap sample. Fixing t is analogous to selecting a best subset, and then using the least squares standard error for that subset.

An approximate closed form estimate may be derived by writing the penalty $\Sigma|\beta_j|$ as $\Sigma \beta_j^2/|\beta_j|$. Hence, at the lasso estimate $\tilde{\beta}$, we may approximate the solution by a

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ridge regression of the form $\beta^* = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{W}^-)^{-1}\mathbf{X}^T\mathbf{y}$ where $\mathbf{W}$ is a diagonal matrix with diagonal elements $|\tilde{\beta}_j|$, $\mathbf{W}^-$ denotes the generalized inverse of $\mathbf{W}$ and λ is chosen so that $\Sigma|\beta_j|^* = t$. The covariance matrix of the estimates may then be approximated by

$$(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{W}^-)^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X} + \lambda\mathbf{X}^-)^{-1}\hat{\sigma}^2, \tag{7}$$

where $\hat{\sigma}^2$ is an estimate of the error variance. A difficulty with this formula is that it gives an estimated variance of 0 for predictors with $\hat{\beta}_j = 0$.

This approximation also suggests an iterated ridge regression algorithm for computing the lasso estimate itself, but this turns out to be quite inefficient. However, it is useful for selection of the lasso parameter t (Section 4).

3. EXAMPLE—PROSTATE CANCER DATA

The prostate cancer data come from a study by Stamey *et al.* (1989) that examined the correlation between the level of prostate specific antigen and a number of clinical measures, in men who were about to receive a radical prostatectomy. The factors were log(cancer volume) (lcavol), log(prostate weight) (lweight), age, log(benign prostatic hyperplasia amount) (lbph), seminal vesicle invasion (svi), log(capsular penetration) (lcp), Gleason score (gleason) and percentage Gleason scores 4 or 5 (pgg45). We fit a linear model to log(prostate specific antigen) (lpsa) after first standardizing the predictors.

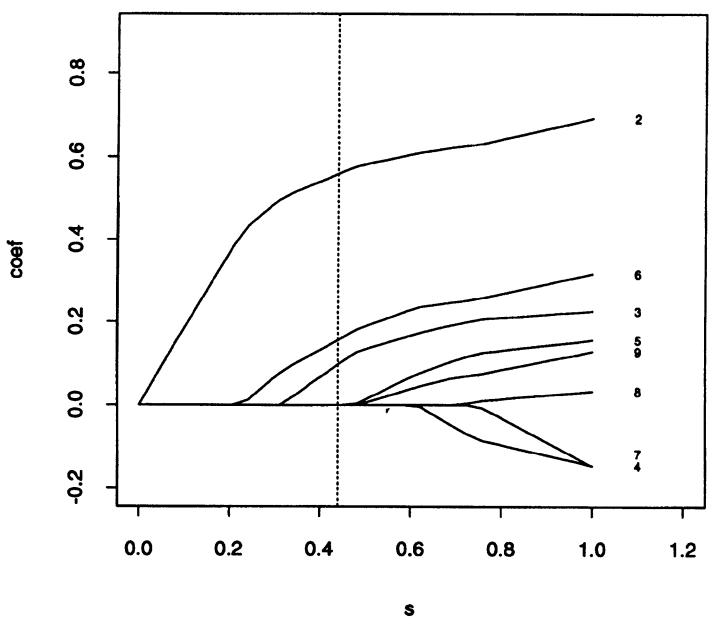


Fig. 5. Lasso shrinkage of coefficients in the prostate cancer example: each curve represents a coefficient (labelled on the right) as a function of the (scaled) lasso parameter $s = t/\Sigma|\beta_j^\circ|$ (the intercept is not plotted); the broken line represents the model for $\hat{s} = 0.44$, selected by generalized cross-validation

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Fig. 5 shows the lasso estimates as a function of standardized bound $s = t/\Sigma|\hat{\beta}_j^0|$. Notice that the absolute value of each coefficient tends to 0 as s goes to 0. In this example, the curves decrease in a monotone fashion to 0, but this does not always happen in general. This lack of monotonicity is shared by ridge regression and subset regression, where for example the best subset of size 5 may not contain the best subset of size 4. The vertical broken line represents the model for $\hat{s} = 0.44$, the optimal value selected by generalized cross-validation. Roughly, this corresponds to keeping just under half of the predictors.

Table 1 shows the results for the full least squares, best subset and lasso procedures. Section 7.1 gives the details of the best subset procedure that was used. The lasso gave non-zero coefficients to lcavol, lweight and svi; subset selection chose the same three predictors. Notice that the coefficients and Z-scores for the selected predictors from subset selection tend to be larger than the full model values: this is common with positively correlated predictors. However, the lasso shows the opposite effect, as it shrinks the coefficients and Z-scores from their full model values.

The standard errors in the penultimate column were estimated by bootstrap resampling of residuals from the full least squares fit. The standard errors were computed by fixing $\hat{s}$ at its optimal value 0.44 for the original data set. Table 2

TABLE 1
Results for the prostate cancer example

Predictor	*Least squares results*			*Subset selection results*			*Lasso results*		
	Coefficient	*Standard error*	*Z-score*	*Coefficient*	*Standard error*	*Z-score*	*Coefficient*	*Standard error*	*Z-score*
1 intcpt	2.48	0.07	34.46	2.48	0.07	34.05	2.48	0.07	35.43
2 lcavol	0.69	0.10	6.68	0.65	0.09	7.39	0.56	0.09	6.22
3 lweight	0.23	0.08	2.67	0.25	0.07	3.39	0.10	0.07	1.43
4 age	−0.15	0.08	−1.76	0.00	0.00	—	0.00	0.01	0.00
5 lbph	0.16	0.08	1.83	0.00	0.00	—	0.00	0.04	0.00
6 svi	0.32	0.10	3.14	0.28	0.09	3.18	0.16	0.09	1.78
7 lcp	−0.15	0.13	−1.16	0.00	0.00	—	0.00	0.03	0.00
8 gleason	0.03	0.11	0.29	0.00	0.00	—	0.00	0.02	0.00
9 pgg45	0.13	0.12	1.02	0.00	0.00	—	0.00	0.03	0.00

TABLE 2
Standard error estimates for the prostate cancer example

Predictor	*Coefficient*	*Bootstrap standard error*		*Standard error approximation (7)*
		Fixed t	*Varying t*	
1 intcpt	2.48	0.07	0.07	0.07
2 lcavol	0.56	0.08	0.10	0.09
3 lweight	0.10	0.06	0.08	0.06
4 age	0.00	0.04	0.05	0.00
5 lbph	0.00	0.04	0.07	0.00
6 svi	0.16	0.09	0.09	0.07
7 lcp	0.00	0.03	0.07	0.00
8 gleason	0.00	0.02	0.05	0.00
9 pgg45	0.00	0.03	0.06	0.00

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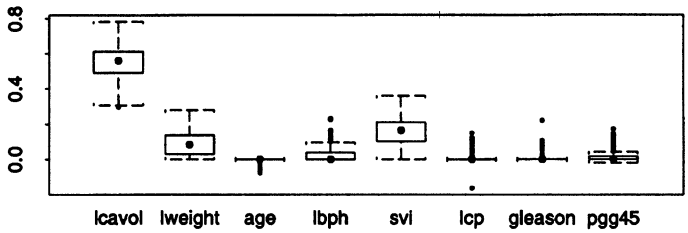


Fig. 6. Box plots of 200 bootstrap values of the lasso coefficient estimates for the eight predictors in the prostate cancer example

compares the ridge approximation formula (7) with the fixed t bootstrap, and the bootstrap in which t was re-estimated for each sample. The ridge formula gives a fairly good approximation to the fixed t bootstrap, except for the zero coefficients. Allowing t to vary incorporates an additional source of variation and hence gives larger standard error estimates. Fig. 6 shows box plots of 200 bootstrap replications of the lasso estimates, with $\hat{s}$ fixed at the estimated value 0.44. The predictors whose estimated coefficient is 0 exhibit skewed bootstrap distributions. The central 90% percentile intervals (fifth and 95th percentiles of the bootstrap distributions) all contained the value 0, with the exceptions of those for lcavol and svi.

4. PREDICTION ERROR AND ESTIMATION OF t

In this section we describe three methods for the estimation of the lasso parameter t: cross-validation, generalized cross-validation and an analytical unbiased estimate of risk. Strictly speaking the first two methods are applicable in the 'X-random' case, where it is assumed that the observations $(\mathbf{X}, Y)$ are drawn from some unknown distribution, and the third method applies to the X-fixed case. However, in real problems there is often no clear distinction between the two scenarios and we might simply choose the most convenient method.

Suppose that

$$Y = \eta(\mathbf{X}) + \epsilon$$

where $E(\epsilon) = 0$ and $\text{var}(\epsilon) = \sigma^2$. The mean-squared error of an estimate $\hat{\eta}(\mathbf{X})$ is defined by

$$\text{ME} = E\{\hat{\eta}(\mathbf{X}) - \eta(\mathbf{X})\}^2,$$

the expected value taken over the joint distribution of $\mathbf{X}$ and Y, with $\hat{\eta}(\mathbf{X})$ fixed. A similar measure is the prediction error of $\hat{\eta}(\mathbf{X})$ given by

$$\text{PE} = E\{Y - \hat{\eta}(\mathbf{X})\}^2 = \text{ME} + \sigma^2. \tag{8}$$

We estimate the prediction error for the lasso procedure by fivefold cross-validation as described (for example) in chapter 17 of Efron and Tibshirani (1993). The lasso is indexed in terms of the normalized parameter $s = t/\Sigma\,\hat{\beta}_j^{\circ}$, and the prediction error is estimated over a grid of values of s from 0 to 1 inclusive. The value $\hat{s}$ yielding the lowest estimated PE is selected.

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Simulation results are reported in terms of ME rather than PE. For the linear models $\eta(\mathbf{X}) = \mathbf{X}\hat{\beta}$ considered in this paper, the mean-squared error has the simple form

$$\text{ME} = (\hat{\beta} - \beta)^{\mathrm{T}} V (\hat{\beta} - \beta)$$

where V is the population covariance matrix of $\mathbf{X}$.

A second method for estimating t may be derived from a linear approximation to the lasso estimate. We write the constraint $\Sigma|\beta_j| \leqslant t$ as $\Sigma\, \beta_j^2/|\beta_j| \leqslant t$. This constraint is equivalent to adding a Lagrangian penalty $\lambda\, \Sigma\, \beta_j^2/|\beta_j|$ to the residual sum of squares, with λ depending on t. Thus we may write the constrained solution $\tilde{\beta}$ as the ridge regression estimator

$$\tilde{\beta} = (\mathbf{X}^{\mathrm{T}}\mathbf{X} + \lambda \mathbf{W}^{-})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y} \tag{9}$$

where $\mathbf{W} = \text{diag}(|\tilde{\beta}_j|)$ and $\mathbf{W}^{-}$ denotes a generalized inverse. Therefore the number of effective parameters in the constrained fit $\tilde{\beta}$ may be approximated by

$$p(t) = \text{tr}\{\mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X} + \lambda \mathbf{W}^{-})^{-1}\mathbf{X}^{\mathrm{T}}\}.$$

Letting $\text{rss}(t)$ be the residual sum of squares for the constrained fit with constraint t, we construct the generalized cross-validation style statistic

$$\text{GCV}(t) = \frac{1}{N}\,\frac{\text{rss}(t)}{\{1 - p(t)/N\}^2}. \tag{10}$$

Finally, we outline a third method based on Stein's unbiased estimate of risk. Suppose that $\mathbf{z}$ is a multivariate normal random vector with mean μ and variance the identity matrix. Let $\hat{\mu}$ be an estimator of μ, and write $\hat{\mu} = \mathbf{z} + \mathbf{g}(\mathbf{z})$ where $\mathbf{g}$ is an almost differential function from R^p to R^p (see definition 1 of Stein (1981)). Then Stein (1981) showed that

$$E_\mu ||\hat{\mu} - \mu||^2 = p + E_\mu\left(||\mathbf{g}(\mathbf{z})||^2 + 2\sum_1^p \mathrm{d}g_i/\mathrm{d}z_i\right). \tag{11}$$

We may apply this result to the lasso estimator (3). Denote the estimated standard error of $\hat{\beta}_j^{\circ}$ by $\hat{\tau} = \hat{\sigma}/\sqrt{N}$, where $\hat{\sigma}^2 = \Sigma(y_i - \hat{y}_i)^2/(N - p)$. Then the $\hat{\beta}_j^{\circ}/\hat{\tau}$ are (conditionally on $\mathbf{X}$) approximately independent standard normal variates, and from equation (11) we may derive the formula

$$R\{\hat{\beta}(\gamma)\} \approx \hat{\tau}^2\left\{p - 2\,\#(j;\, |\hat{\beta}_j^{\circ}/\hat{\tau}| < \gamma) + \sum_{j=1}^{p} \max(|\hat{\beta}_j^{\circ}/\hat{\tau}|,\, \gamma)^2\right\}$$

as an approximately unbiased estimate of the risk or mean-square error $E\{\hat{\beta}(\gamma) - \beta\}^2$, where $\hat{\beta}_j(\gamma) = \text{sign}(\hat{\beta}_j^{\circ})(|\hat{\beta}_j^{\circ}/\hat{\tau}| - \gamma)^{+}$. Donoho and Johnstone (1994) gave a similar formula in the function estimation setting. Hence an estimate of γ can be obtained as the minimizer of $R\{\hat{\beta}(\gamma)\}$:

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$$\hat{\gamma} = \arg\min_{\gamma \geqslant 0}[R\{\hat{\beta}(\gamma)\}].$$

From this we obtain an estimate of the lasso parameter t:

$$\hat{t} = \sum (|\hat{\beta}_j^{\circ}| - \hat{\gamma})^+.$$

Although the derivation of $\hat{t}$ assumes an orthogonal design, we may still try to use it in the usual non-orthogonal setting. Since the predictors have been standardized, the optimal value of t is roughly a function of the overall signal-to-noise ratio in the data, and it should be relatively insensitive to the covariance of $\mathbf{X}$. (In contrast, the form of the lasso estimator *is* sensitive to the covariance and we need to account for it properly.)

The simulated examples in Section 7.2 suggest that this method gives a useful estimate of t. But we can offer only a heuristic argument in favour of it. Suppose that $\mathbf{X}^{\mathrm{T}}\mathbf{X} = \mathbf{V}$ and let $\mathbf{Z} = \mathbf{X}\mathbf{V}^{-1/2}$, $\theta = \beta\mathbf{V}^{-1/2}$. Since the columns of $\mathbf{X}$ are standardized, the region $\Sigma|\theta_j| \leqslant t$ differs from the region $\Sigma|\beta_j| \leqslant t$ in shape but has roughly the same-sized marginal projections. Therefore the optimal value of $\hat{t}$ should be about the same in each instance.

Finally, note that the Stein method enjoys a significant computational advantage over the cross-validation-based estimate of t. In our experiments we optimized over a grid of 15 values of the lasso parameter t and used fivefold cross-validation. As a result, the cross-validation approach required 75 applications of the model optimization procedure of Section 6 whereas the Stein method required only one. The requirements of the generalized cross-validation approach are intermediate between the two, requiring one application of the optimization procedure per grid point.

5. LASSO AS BAYES ESTIMATE

The lasso constraint $\Sigma|\beta_j| \leqslant t$ is equivalent to the addition of a penalty term $\lambda\,\Sigma|\beta_j|$ to the residual sum of squares (see Murray *et al.* (1981), chapter 5). Now $|\beta_j|$ is proportional to the (minus) log-density of the double-exponential distribution. As a result we can derive the lasso estimate as the Bayes posterior mode under independent double-exponential priors for the β_js,

$$f(\beta_j) = \frac{1}{2\tau}\exp\left(-\frac{|\beta_j|}{\tau}\right)$$

with $\tau = 1/\lambda$.

Fig. 7 shows the double-exponential density (full curve) and the normal density (broken curve); the latter is the implicit prior used by ridge regression. Notice how the double-exponential density puts more mass near 0 and in the tails. This reflects the greater tendency of the lasso to produce estimates that are either large or 0.

6. ALGORITHMS FOR FINDING LASSO SOLUTIONS

We fix $t \geqslant 0$. Problem (1) can be expressed as a least squares problem with 2^p inequality constraints, corresponding to the 2^p different possible signs for the β_js.

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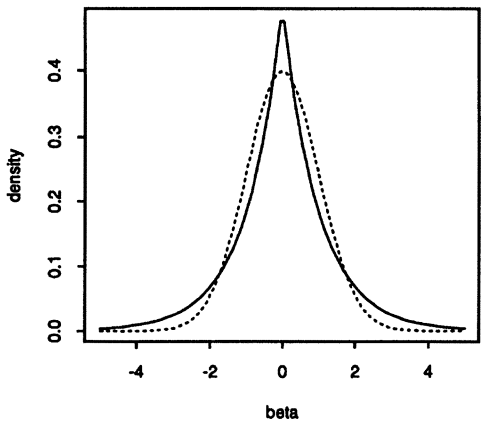


Fig. 7. Double-exponential density (———) and normal density (- - - - -): the former is the implicit prior used by the lasso; the latter by ridge regression

Lawson and Hansen (1974) provided the ingredients for a procedure which solves the linear least squares problem subject to a general linear inequality constraint $G\beta \leqslant \mathbf{h}$. Here G is an $m \times p$ matrix, corresponding to m linear inequality constraints on the p-vector β. For our problem, however, $m = 2^p$ may be very large so that direct application of this procedure is not practical. However, the problem can be solved by introducing the inequality constraints sequentially, seeking a feasible solution satisfying the so-called Kuhn–Tucker conditions (Lawson and Hansen, 1974). We outline the procedure below.

Let $g(\beta) = \Sigma_{i=1}^{N} (y_i - \Sigma_j \beta_j x_{ij})^2$, and let δ_i, $i = 1, 2, \ldots, 2^p$ be the p-tuples of the form $(\pm 1, \pm 1, \ldots, \pm 1)$. Then the condition $\Sigma|\beta_j| \leqslant t$ is equivalent to $\delta_i^{\mathrm{T}}\beta \leqslant t$ for all i. For a given β, let $E = \{i: \delta_i^{\mathrm{T}}\beta = t\}$ and $S = \{i: \delta_i^{\mathrm{T}}\beta < t\}$. The set E is the equality set, corresponding to those constraints which are exactly met, whereas S is the slack set, corresponding to those constraints for which equality does not hold. Denote by G_E the matrix whose rows are δ_i for $i \in E$. Let $\mathbf{1}$ be a vector of 1s of length equal to the number of rows of G_E.

The following algorithm starts with $E = \{i_0\}$ where $\delta_{i_0} = \operatorname{sign}(\hat{\beta})$, $\hat{\beta}$ being the overall least squares estimate. It solves the least squares problem subject to $\delta_{i_0}^{\mathrm{T}}\beta \leqslant t$ and then checks whether $\Sigma|\beta_j| \leqslant t$. If so, the computation is complete; if not, the violated constraint is added to E and the process is continued until $\Sigma|\beta_j| \leqslant t$.

Here is an outline of the algorithm.

(a) Start with $E = \{i_0\}$ where $\delta_{i_0} = \operatorname{sign}(\hat{\beta}^{\circ})$, $\hat{\beta}^{\circ}$ being the overall least squares estimate.
(b) Find $\hat{\beta}$ to minimize $g(\beta)$ subject to $G_E\beta \leqslant t\mathbf{1}$.
(c) While $\{\Sigma|\hat{\beta}_j| > t\}$,
(d) add i to the set E where $\delta_i = \operatorname{sign}(\hat{\beta})$. Find $\hat{\beta}$ to minimize $g(\beta)$ subject to $G_E\beta \leqslant t\mathbf{1}$.

This procedure must always converge in a finite number of steps since one element is added to the set E at each step, and there is a total of 2^p elements. The final iterate

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TABLE 3
Results for example 1†

Method	*Median mean-squared error*	*Average no. of 0 coefficients*	*Average ŝ*
Least squares	2.79 (0.12)	0.0	—
Lasso (cross-validation)	2.43 (0.14)	3.3	0.63 (0.01)
Lasso (Stein)	2.07 (0.10)	2.6	0.69 (0.02)
Lasso (generalized cross-validation)	1.93 (0.09)	2.4	0.73 (0.01)
Garotte	2.29 (0.16)	3.9	—
Best subset selection	2.44 (0.16)	4.8	—
Ridge regression	3.21 (0.12)	0.0	—

†Standard errors are given in parentheses.

is a solution to the original problem since the Kuhn–Tucker conditions are satisfied for the sets E and S at convergence.

A modification of this procedure removes elements from E in step (d) for which the equality constraint is not satisfied. This is more efficient but it is not clear how to establish its convergence.

The fact that the algorithm must stop after at most 2^p iterations is of little comfort if p is large. In practice we have found that the average number of iterations required is in the range $(0.5p, 0.75p)$ and is therefore quite acceptable for practical purposes.

A completely different algorithm for this problem was suggested by David Gay. We write each β_j as $\beta_j^+ - \beta_j^-$, where β_j^+ and β_j^- are non-negative. Then we solve the least squares problem with the constraints $\beta_j^+ \geqslant 0$, $\beta_j^- \geqslant 0$ and $\Sigma\, \beta_j^+ + \Sigma_j\, \beta_j^- \leqslant t$. In this way we transform the original problem (p variables, 2^p constraints) to a new problem with more variables ($2p$) but fewer constraints ($2p + 1$). One can show that this new problem has the same solution as the original problem.

Standard quadratic programming techniques can be applied, with the convergence assured in $2p + 1$ steps. We have not extensively compared these two algorithms but in examples have found that the second algorithm is usually (but not always) a little faster than the first.

7. SIMULATIONS

7.1. *Outline*

In the following examples, we compare the full least squares estimates with the lasso, the non-negative garotte, best subset selection and ridge regression. We used fivefold cross-validation to estimate the regularization parameter in each case. For best subset selection, we used the 'leaps' procedure in the S language, with fivefold cross-validation to estimate the best subset size. This procedure is described and studied in Breiman and Spector (1992) who recommended fivefold or tenfold cross-validation for use in practice.

For completeness, here are the details of the cross-validation procedure. The best subsets of each size are first found for the original data set: call these $S_0, S_1, \ldots, S_p$. (S_0 represents the null nodel; since $\bar{y} = 0$ the fitted values are 0 for this model.) Denote the full training set by T, and the cross-validation training and test sets by $T - T^v$ and T^v, for $v = 1, 2, \ldots, 5$. For each cross-validation fold v, we find the best

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TABLE 4
Most frequent models selected by the lasso (generalized cross-validation) in example 1

Model	*Proportion*
1245678	0.055
123456	0.050
1258	0.045
1245	0.045
13 others	
125 (and 5 others)	0.025

subsets of each size for the data $T - T^v$: call these S_0^v, S_1^v, . . ., S_p^v. Let $\mathrm{PE}^v(J)$ be the prediction error when S_J^v is applied to the test data T^v, and form the estimate

$$\mathrm{PE}(J) = \frac{1}{5}\sum_{v=1}^{5} \mathrm{PE}^v(J). \tag{12}$$

We find the $\hat{J}$ that minimizes $\mathrm{PE}(J)$ and our selected model is $S_{\hat{J}}$. This is not the same as estimating the prediction error of the fixed models S_0, S_1, . . ., S_p and then choosing the one with the smallest prediction error. This latter procedure is described in Zhang (1993) and Shao (1992), and can lead to inconsistent model selection unless the cross-validation test set T^v grows at an appropriate asymptotic rate.

7.2. *Example 1*

In this example we simulated 50 data sets consisting of 20 observations from the model

$$y = \beta^{\mathrm{T}}\mathbf{x} + \sigma\epsilon,$$

where $\beta = (3, 1.5, 0, 0, 2, 0, 0, 0)^{\mathrm{T}}$ and ϵ is standard normal. The correlation between x_i and x_j was $\rho^{|i-j|}$ with $\rho = 0.5$. We set $\sigma = 3$, and this gave a signal-to-noise ratio of approximately 5.7. Table 3 shows the mean-squared errors over 200 simulations from this model. The lasso performs the best, followed by the garotte and ridge regression.

Estimation of the lasso parameter by generalized cross-validation seems to perform best, a trend that we find is consistent through all our examples. Subset

TABLE 5
Most frequent models selected by all-subsets regression in example 1

Model	Proportion
125	0.240
15	0.200
1	0.095
1257	0.040

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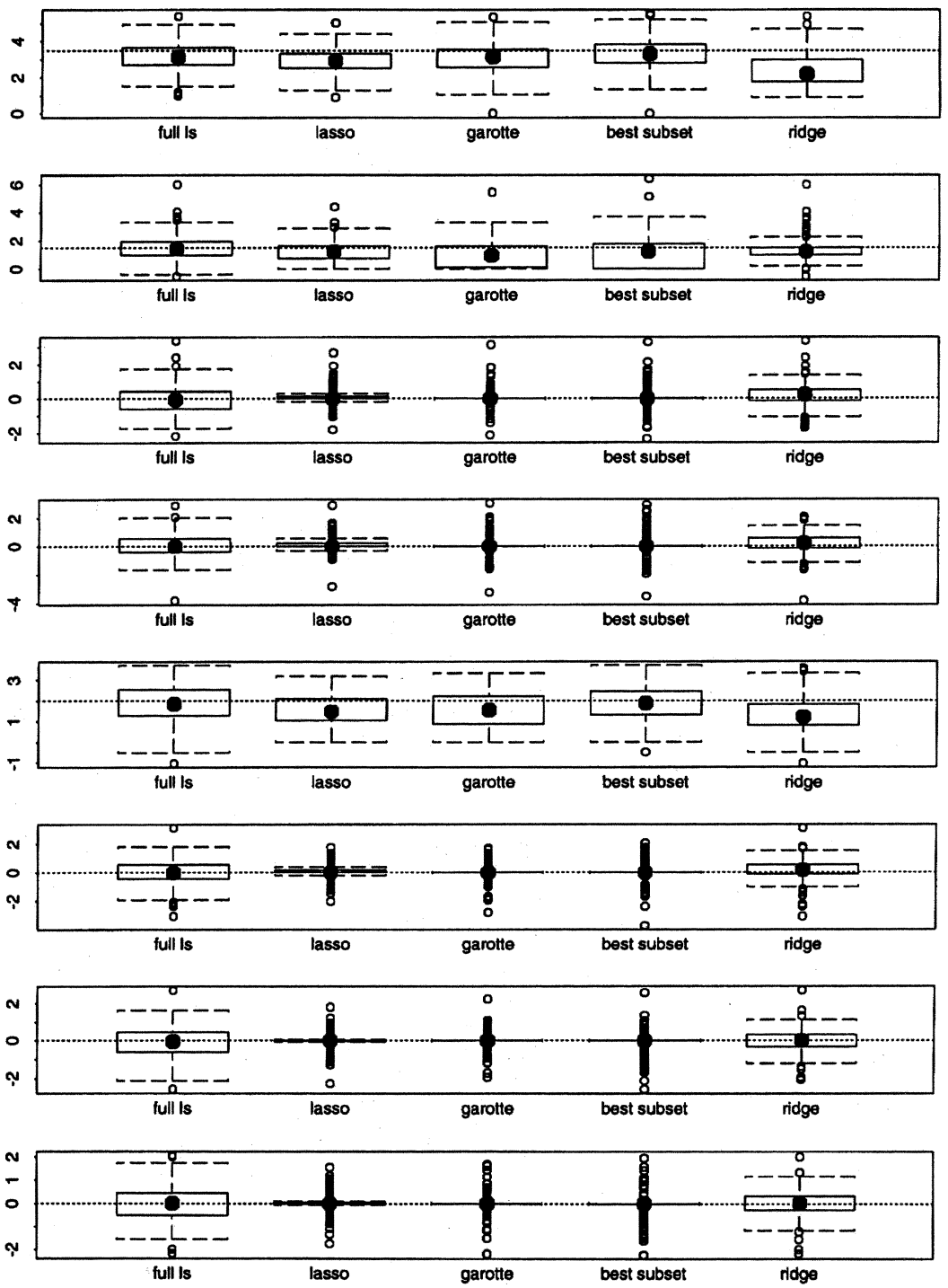


Fig. 8. Estimates for the eight coefficients in example 1, excluding the intercept: ·····, true coefficients

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TABLE 6
Results for example 2†

Method	*Median mean-squared error*	*Average no. of 0 coefficients*	*Average $\hat{s}$*
Least squares	6.50 (0.64)	0.0	—
Lasso (cross-validation)	5.30 (0.45)	3.0	0.50 (0.03)
Lasso (Stein)	5.85 (0.36)	2.7	0.55 (0.03)
Lasso (generalized cross-validation)	4.87 (0.35)	2.3	0.69 (0.23)
Garotte	7.40 (0.48)	4.3	—
Subset selection	9.05 (0.78)	5.2	—
Ridge regression	2.30 (0.22)	0.0	—

†Standard errors are given in parentheses.

selection picks approximately the correct number of zero coefficients (5), but suffers from too much variability as shown in the box plots of Fig. 8.

Table 4 shows the five most frequent models (non-zero coefficients) selected by the lasso (with generalized cross-validation): although the correct model (1, 2, 5) was chosen only 2.5% of the time, the selected model contained (1, 2, 5) 95.5% of the time. The most frequent models selected by subset regression are shown in Table 5. The correct model is chosen more often (24% of the time), but subset selection can also underfit: the selected model contained (1, 2, 5) only 53.5% of the time.

7.3. *Example 2*

The second example is the same as example 1, but with $\beta_j = 0.85$, $\forall j$ and $\sigma = 3$; the signal-to-noise ratio was approximately 1.8. The results in Table 6 show that ridge regression does the best by a good margin, with the lasso being the only other method to outperform the full least squares estimate.

7.4. *Example 3*

For example 3 we chose a set-up that should be well suited for subset selection. The model is the same as example 1, but with $\beta = (5, 0, 0, 0, 0, 0, 0, 0)$ and $\sigma = 2$ so that the signal-to-noise ratio was about 7.

The results in Table 7 show that the garotte and subset selection perform the best,

TABLE 7
Results for example 3†

Method	*Median mean-squared error*	*Average no. of 0 coefficients*	*Average $\hat{s}$*
Least squares	2.89 (0.04)	0.0	—
Lasso (cross-validation)	0.89 (0.01)	3.0	0.50 (0.03)
Lasso (Stein)	1.26 (0.02)	2.6	0.70 (0.01)
Lasso (generalized cross-validation)	1.02 (0.02)	3.9	0.63 (0.04)
Garotte	0.52 (0.01)	5.5	—
Subset selection	0.64 (0.02)	6.3	—
Ridge regression	3.53 (0.05)	0.0	—

†Standard errors are given in parentheses.

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TABLE 8
Results for example 4†

Method	*Median mean-squared error*	*Average no. of 0 coefficients*	*Average $\hat{s}$*
Least squares	137.3 (7.3)	0.0	—
Lasso (Stein)	80.2 (4.9)	14.4	0.55 (0.02)
Lasso (generalized cross-validation)	64.9 (2.3)	13.6	0.60 (0.88)
Garotte	94.8 (3.2)	22.9	—
Ridge regression	57.4 (1.4)	0.0	—

†Standard errors are given in parentheses.

followed closely by the lasso. Ridge regression does poorly and has a higher mean-squared error than do the full least squares estimates.

7.5. *Example 4*

In this example we examine the performance of the lasso in a bigger model. We simulated 50 data sets each having 100 observations and 40 variables (note that best subsets regression is generally considered impractical for $p > 30$). We defined predictors $x_{ij} = z_{ij} + z_i$ where z_{ij} and z_i are independent standard normal variates. This induced a pairwise correlation of 0.5 among the predictors. The coefficient vector was $\beta = (0, 0, \ldots, 0, 2, 2, \ldots, 2, 0, 0, \ldots, 0, 2, 2, \ldots, 2)$, there being 10 repeats in each block. Finally we defined $y = \beta^T\mathbf{x} + 15\epsilon$ where ϵ was standard normal. This produced a signal-to-noise ratio of roughly 9. The results in Table 8 show that the ridge regression performs the best, with the lasso (generalized cross-validation) a close second.

The average value of the lasso coefficients in each of the four blocks of 10 were 0.50 (0.06), 0.92 (0.07), 1.56 (0.08) and 2.33 (0.09). Although the lasso only produced 14.4 zero coefficients on average, the average value of $\hat{s}$ (0.55) was close to the true proportion of 0s (0.5).

8. APPLICATION TO GENERALIZED REGRESSION MODELS

The lasso can be applied to many other models: for example Tibshirani (1994) described an application to the proportional hazards model. Here we briefly explore the application to generalized regression models.

Consider any model indexed by a vector parameter β, for which estimation is carried out by maximization of a function $l(\beta)$; this may be a log-likelihood function or some other measure of fit. To apply the lasso, we maximize $l(\beta)$ under the constraint $\Sigma|\beta_j| \leqslant t$. It might be possible to carry out this maximization by a general (non-quadratic) programming procedure. Instead, we consider here models for which a quadratic approximation to $l(\beta)$ leads to an iteratively reweighted least squares (IRLS) procedure for computation of β. Such a procedure is equivalent to a Newton–Raphson algorithm. Using this approach, we can solve the constrained problem by iterative application of the lasso algorithm, within an IRLS loop. Convergence of this procedure is not ensured in general, but in our limited experience it has behaved quite well.

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8.1. *Logistic Regression*

For illustration we applied the lasso to the logistic regression model for binary data. We used the kyphosis data, analysed in Hastie and Tibshirani (1990), chapter 10. The response is kyphosis ($0 \equiv$ absent, $1 \equiv$ present); the predictors $x_1 \equiv$ age, $x_2 \equiv$ number of vertebrae levels and $x_3 \equiv$ starting vertebrae level. There are 83 observations. Since the predictor effects are known to be non-linear, we included squared terms in the model after centring each of the variables. Finally, the columns of the data matrix were standardized.

The linear logistic fitted model is

$$-2.64 + 0.83x_1 + 0.77x_2 - 2.28x_3 - 1.55x_1^2 + 0.03x_2^2 - 1.17x_3^2.$$

Backward stepwise deletion, based on Akaike's information criterion, dropped the x_2^2-term and produced the model

$$-2.64 + 0.84x_1 + 0.80x_2 - 2.28x_3 - 1.54x_1^2 - 1.16x_3^2.$$

The lasso chose $\hat{s} = 0.33$, giving the model

$$-1.42 + 0.03x_1 + 0.31x_2 - 0.48x_3 - 0.28x_1^2.$$

Convergence, defined as the $||\hat{\beta}^{\text{new}} - \hat{\beta}^{\text{old}}||^2 < 10^{-6}$, was obtained in five iterations.

9. SOME FURTHER EXTENSIONS

We are currently exploring two quite different applications of the lasso idea. One application is to tree-based models, as reported in LeBlanc and Tibshirani (1994). Rather than prune a large tree as in the classification and regression tree approach of Breiman *et al.* (1984), we use the lasso idea to shrink it. This involves a constrained least squares operation much like that in this paper, with the parameters being the mean contrasts at each node. A further set of constraints is needed to ensure that the shrunken model is a tree. Results reported in LeBlanc and Tibshirani (1994) suggest that the shrinkage procedure gives more accurate trees than pruning, while still producing interpretable subtrees.

A different application is to the multivariate adaptive regression splines (MARS) proposal of Friedman (1991). The MARS approach is an adaptive procedure that builds a regression surface by sum of products of piecewise linear basis functions of the individual regressors. The MARS algorithm builds a model that typically includes basis functions representing main effects and interactions of high order. Give the adaptively chosen bases, the MARS fit is simply a linear regression onto these bases. A backward stepwise procedure is then applied to eliminate less important terms.

In on-going work with Trevor Hastie, we are developing a special lasso-type algorithm to grow and prune a MARS model dynamically. Hopefully this will produce more accurate MARS models which also are interpretable.

The lasso idea can also be applied to ill-posed problems, in which the predictor matrix is not full rank. Chen and Donoho (1994) reported some encouraging results for the use of lasso-style constraints in the context of function estimation via wavelets.

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10. RESULTS ON SOFT THRESHOLDING

Consider the special case of an orthonormal design $\mathbf{X}^T\mathbf{X} = \mathbf{I}$. Then the lasso estimate has the form

$$\hat{\beta}_j = \text{sign}(\hat{\beta}_j^o)(|\hat{\beta}_j^o| - \gamma)^+. \tag{13}$$

This is called a 'soft threshold' estimator by Donoho and Johnstone (1994); they applied this estimator to the coefficients of a wavelet transform of a function measured with noise. They then backtransformed to obtain a smooth estimate of the function. Donoho and Johnstone proved many optimality results for the soft threshold estimator and then translated these results into optimality results for function estimation.

Our interest here is not in function estimation but in the coefficients themselves. We give one of Donoho and Johnstone's results here. It shows that asymptotically the soft threshold estimator (lasso) comes as close as subset selection to the performance of an ideal subset selector — one that uses information about the actual parameters.

Suppose that

$$y_i = \beta \mathbf{x}^i + \epsilon_i$$

where $\epsilon_i \sim N(0, \sigma^2)$ and the design matrix is orthonormal. Then we can write

$$\hat{\beta}_j^o = \beta_j + \sigma z_j \tag{14}$$

where $z_j \sim N(0, \sigma^2)$.

We consider estimation of β under squared error loss, with risk

$$R(\hat{\beta}, \beta) = E||\hat{\beta} - \beta||^2.$$

Consider the family of diagonal linear projections

$$T_{\text{DP}}(\hat{\beta}^o, \delta) = (\delta_j \hat{\beta}_j)_{j=1}^p \qquad \delta \in \{0, 1\}. \tag{15}$$

This estimator either keeps or kills a parameter $\hat{\beta}_j^o$, i.e. it does subset selection. Now we incur a risk of σ^2 if we use $\hat{\beta}_j^o$, and β_j^2 if we use an estimate of 0 instead. Hence the ideal choice of δ_j is $I(|\beta_j| > \sigma)$, i.e. we keep only those predictors whose true coefficient is larger than the noise level. Call the risk of this estimator R_{DP}: of course this estimator cannot be constructed since the β_j are unknown. Hence R_{DP} is a lower bound on the risk that we can hope to attain.

Donoho and Johnstone (1994) proved that the hard threshold (subset selection) estimator $\tilde{\beta}_j = \hat{\beta}_j^o I(|\hat{\beta}_j^o| > \gamma)$ has risk

$$R(\tilde{\beta}, \beta) \leqslant (2 \log p + 1)(\sigma^2 + R_{\text{DP}}). \tag{16}$$

Here γ is chosen as $\sigma(2 \log n)^{1/2}$, the choice giving smallest asymptotic risk. They also showed that the soft threshold estimator (13) with $\gamma = \sigma(2 \log n)^{1/2}$ achieves the same asymptotic rate.

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These results lend some support to the potential utility of the lasso in linear models. However, the important differences between the various approaches tend to occur for correlated predictors, and theoretical results such as those given here seem to be more difficult to obtain in that case.

11. DISCUSSION

In this paper we have proposed a new method (the lasso) for shrinkage and selection for regression and generalized regression problems. The lasso does not focus on subsets but rather defines a continuous shrinking operation that can produce coefficients that are exactly 0. We have presented some evidence in this paper that suggests that the lasso is a worthy competitor to subset selection and ridge regression. We examined the relative merits of the methods in three different scenarios:

(a) *small number of large effects*—subset selection does best here, the lasso not quite as well and ridge regression does quite poorly;
(b) *small to moderate number of moderate-sized effects*—the lasso does best, followed by ridge regression and then subset selection;
(c) *large number of small effects*—ridge regression does best by a good margin, followed by the lasso and then subset selection.

Breiman's garotte does a little better than the lasso in the first scenario, and a little worse in the second two scenarios. These results refer to prediction accuracy. Subset selection, the lasso and the garotte have the further advantage (compared with ridge regression) of producing interpretable submodels.

There are many other ways to carry out subset selection or regularization in least squares regression. The literature is increasing far too fast to attempt to summarize it in this short space so we mention only a few recent developments. Computational advances have led to some interesting proposals, such as the Gibbs sampling approach of George and McCulloch (1993). They set up a hierarchical Bayes model and then used the Gibbs sampler to simulate a large collection of subset models from the posterior distribution. This allows the data analyst to examine the subset models with highest posterior probability and can be carried out in large problems.

Frank and Friedman (1993) discuss a generalization of ridge regression and subset selection, through the addition of a penalty of the form $\lambda \Sigma_j |\beta_j|^q$ to the residual sum of squares. This is equivalent to a constraint of the form $\Sigma_j |\beta_j|^q \leqslant t$; they called this the 'bridge'. The lasso corresponds to $q = 1$. They suggested that joint estimation of the β_js and q might be an effective strategy but do not report any results.

Fig. 9 depicts the situation in two dimensions. Subset selection corresponds to $q \to 0$. The value $q = 1$ has the advantage of being closer to subset selection than is ridge regression ($q = 2$) and is also the smallest value of q giving a convex region. Furthermore, the linear boundaries for $q = 1$ are convenient for optimization.

The encouraging results reported here suggest that absolute value constraints might prove to be useful in a wide variety of statistical estimation problems. Further study is needed to investigate these possibilities.

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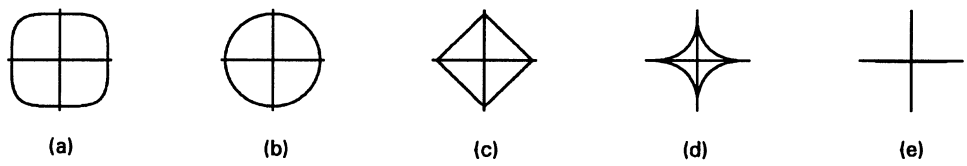


Fig. 9. Contours of constant value of $\Sigma_j|\beta_j|^q$ for given values of q: (a) $q = 4$; (b) $q = 2$; (c) $q = 1$; (d) $q = 0.5$; (e) $q = 0.1$

12. SOFTWARE

Public domain and S-PLUS language functions for the lasso are available at the Statlib archive at Carnegie Mellon University. There are functions for linear models, generalized linear models and the proportional hazards model. To obtain them, use file transfer protocol to lib.stat.cmu.edu and retrieve the file S/lasso, or send an electronic mail message to statlib@lib.stat.cmu.edu with the message send lasso from S.

ACKNOWLEDGEMENTS

I would like to thank Leo Breiman for sharing his garotte paper with me before publication, Michael Carter for assistance with the algorithm of Section 6 and David Andrews for producing Fig. 3 in MATHEMATICA. I would also like to acknowledge enjoyable and fruitful discussions with David Andrews, Shaobeng Chen, Jerome Friedman, David Gay, Trevor Hastie, Geoff Hinton, Iain Johnstone, Stephanie Land, Michael Leblanc, Brenda MacGibbon, Stephen Stigler and Margaret Wright. Comments by the Editor and a referee led to substantial improvements in the manuscript. This work was supported by a grant from the Natural Sciences and Engineering Research Council of Canada.

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